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ORIGINAL ARTICLE

Dapagliflozin for inpatient hyperglycemia in cardiac surgery patients with type 2 diabetes: randomised controlled trial (Dapa-Hospital trial)

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Abstract

Aims To examine the efficacy and safety of dapagliflozin in the treatment of hyperglycemia in cardiac surgery patients with type 2 diabetes (T2D).

Methods Cardiac surgery patients with T2D (n = 250) were randomly assigned (1:1) to receive dapagliflozin plus basal-bolus insulin (DAPA group) or basal-bolus insulin alone (INSULIN group) in the early postoperative period. The primary outcome was mean difference in daily blood glucose (BG) concentrations between groups. The major safety outcomes were the occurrence of severe ketonemia/diabetic ketoacidosis (DKA) and hypoglycemia. All analyses were performed according to the intention-to-treat principle.

Results The median age of the patients was 61 years (range, 55–61), and 219 (87.6%) were men. Overall, the randomization blood glucose was 165 mg/dL (SD, 37) and glycated hemoglobin was 7.7% (SD, 1.4). There were no differences in mean daily BG concentrations (149 vs. 150 mg/dL), mean percentage of readings within target BG of 70–180 mg/dL (82.7% vs. 82.5%), total daily insulin dose (mean, 39 vs. 40 units/day), number of daily insulin injections (median, 3.9 vs. 4), length of hospital stay (median, 10 vs. 10 days), or hospital complications (21.6% vs. 24.8%) between the DAPA and INSULIN groups. The mean plasma ketone levels were significantly higher in the DAPA group than in the INSULIN group at day 3 (0.71 vs. 0.30 mmol/L) and day 5 (0.42 vs. 0.19 mmol/L) of randomization. Six patients in the DAPA group developed severe ketonemia, but no patient developed DKA. There were no differences in the proportion of patients with BG < 70 mg/dL (9.6% vs. 7.2%) between the two groups.

Conclusion Dapagliflozin complementary to basal-bolus insulin does not improve glycemia further over and above the basal-bolus insulin alone in hospitalized cardiac surgery patients. Dapagliflozin significantly increases plasma ketones levels. Safety of dapagliflozin in hospitalized patients needs further investigation.

Trial registration ClinicalTrials.gov NCT05457933.

Keywords Dapagliflozin · SGLT2 inhibitors · Inpatient hyperglycemia · Inpatient management · Hypoglycemia · CGMS

Introduction

The management of hyperglycemia in noncritically ill, hospitalized patients with diabetes mellitus is mainly based on insulin therapy [1, 2]. This usually consists of one dose of long-acting basal insulin and three doses of rapid-acting premeal bolus insulins (basal-bolus insulin). Basal-bolus insulin therapy is, however, labor intensive, requiring multiple insulin injections per day and multiple daily blood glucose checks. The use of oral antidiabetic medications has generally not been recommended for patients admitted to the hospital. This is because of the lack of safety and efficacy

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data, and concerns about hypoglycemia. Oral medication usually has a slow onset of action that might preclude daily dose adjustments and have considerable interactions with concomitantly administered drugs. Oral antidiabetic medications are also withheld during hospitalization because of several safety concerns related to altered pharmacokinetics in cases of end-stage organ disease, such as renal or liver failure. Dipeptidyl peptidase-4 inhibitors (sitagliptin and linagliptin) have been studied for the treatment of hospitalized patients in the noncritical care setting. Complementary to insulin therapy, these drugs improved glycemia and were found to be safe [3–6]. Sodium-glucose cotransporter type 2 (SGLT2) inhibitors are another class of oral glucose-lowering medication that is increasingly being used in patients with T2D, due to multiple pleiotropic effects [7]. These drugs reduce cardiovascular mortality, especially by reducing the risk of heart failure, and also improve renal outcomes [7]. Recently, two randomized controlled trials demonstrated improvement in several cardiac outcomes when SGLT2 inhibitors (empagliflozin and dapagliflozin) were initiated in patients admitted for acute heart failure, with or without diabetes [8, 9]. These studies did not reveal any significant safety issue. However, these studies did not evaluate the glycemic efficacy of SGLT2 inhibitors. Accordingly, this study aimed to examine the efficacy (blood glucose control) and safety (hypoglycemia, severe ketonemia/DKA, hypoglycemia, acute kidney injury) of dapagliflozin for the treatment of hyperglycemia in cardiac surgery patients with T2D.

Material and methods

Participants and setting

The Dapa-Hospital trial was an investigator-initiated, prospective, open-label, treat-to-target, randomized controlled study in individuals with T2D admitted for cardiac surgery (coronary artery bypass grafting, valvular heart surgery or a combination of the two surgeries). This trial was conducted at Medanta The Medicity Hospital, which is a large-volume tertiary care center in North India. The trial was conducted in accordance with the Declaration of Helsinki and the International Conference on Harmonisation Guidelines for Good Clinical Practice. The clinical trial protocol and consent form were approved by the Medanta Institutional Review Board and the Ethics Committee. All the participants provided written, informed consent before participation. The trial is registered with ClinicalTrials.gov, number NCT05457933.

Eligibility criteria

All consecutive patients admitted for cardiac surgery were screened for the study. Patients with T2D who were admitted for cardiac surgery were deemed eligible for the study. Patients with chronic kidney disease (eGFR < 30 ml/min/1.73^2) and those requiring < 12 units of total insulin dose at randomization were excluded from the study. Patients with a body mass index (BMI) less than 18.0 kg/m^2 or a history of recurrent urinary tract infections were also excluded. See supplementary material, Research design and methods.

Baseline assessment

All the study participants underwent a baseline assessment at hospital admission, including anthropometry (weight, height), detailed medical history (especially current drug history, history of urinary tract infections) and physical examination. All participants underwent blood investigations, including glycaemic variables (admission random plasma glucose and HbA_{1c}), liver function tests (total bilirubin, albumin, aspartate aminotransferase, alanine aminotransferase, gamma-glutamyl transpeptidase, alkaline phosphatase), kidney function tests (serum urea, creatinine, uric acid), electrolytes (sodium, potassium), lipid profile (total cholesterol, triacylglycerol, low density lipoprotein, high density lipoprotein), complete blood counts (haemoglobin, total leucocyte count, haematocrit, platelet count), and hormones (thyrotropin, free thyroxine). All patients underwent transthoracic echocardiography at hospital admission. Left ventricular ejection fraction (LVEF) was defined as normal (LVEF $\geq 50\%$), mild dysfunction (LVEF 40–49%), moderate dysfunction (LVEF 30–39%) and severe dysfunction (LVEF $< 30\%$). The patient was considered to have arterial hypertension if the sitting blood pressure was $\geq 140/90$ mmHg or if anti-hypertensive medication was used. Dyslipidemia was defined as the use of lipid-lowering medication or when the serum low-density lipoprotein (LDL) cholesterol level was > 100 mg/dL. Hypothyroidism was defined as the use of thyroxine or if the serum TSH level was > 10 mIU/mL on admission. Obesity was defined as a body mass index (BMI) ≥ 25 kg/m^2 in accordance with the World Health Organization (WHO) Asia Pacific guidelines.

Randomization and allocation concealment

Resident doctors (PK, AS) and a diabetes educator (MM) randomised the participants into either the DAPA or the INSULIN group. The randomization was performed using a predefined computer-generated number with a 1:1 allocation

that was concealed using serially numbered, opaque, sealed envelopes (prepared by statisticians, MKS, PS). Participants were randomly assigned to receive either dapagliflozin 10 mg (Tablet Oxra 10 mg, Sun Pharmaceutical Industries Ltd, India) daily plus basal-bolus insulin or basal-bolus insulin regimen. The basal-bolus insulin regimen consisted of the rapid-acting insulin analogue lispro (Humalog, Eli Lilly and company) as boluses and glargine U300 (Toujeo; Sanofi) as basal insulin in all patients. Investigators and participants were not masked to the treatment allocation (open-label).

Procedures

Patients were managed for glycemic control by the investigators (MSK, SKM, PK, JSW, HKG, RJ, PK) throughout the study period. Patients were also directly observed for glucose control, insulin dose adjustment, occurrence of hypoglycemia and other relevant complications twice a day (MSK, MM, PK, AS). Participants were treated with a basal-bolus insulin regimen approach, as previously reported (supplementary Table 1) [10]. The study design is shown in Supplementary Fig. 1. In brief, all patients continued antidiabetic medications until the night before surgery. During surgery, all patients received intravenous insulin infusion as per the hospital intravenous insulin infusion protocol (Supplementary material, perioperative insulin protocol) [10]. On the day of transition (day 2 of surgery) from intravenous insulin infusion to subcutaneous insulin, all patients were randomized to receive either dapagliflozin 10 mg daily plus a basal-bolus regimen or a basal-bolus insulin regimen. Insulins glargine U300 as basal and lispro as boluses before meals were used for all patients. The first dose of basal insulin was given at 11:00 am, and infusion was continued until lunchtime (1:00 pm). At the lunch time, the first bolus dose was administered, and insulin infusion was discontinued. Insulin doses were adjusted as per protocol shown in Supplementary Table 1. Patients treated with dapagliflozin received a single daily dose of 10 mg in the morning.

Glucose and ketone monitoring

Admission random blood glucose and glycated hemoglobin (HbA1c) were assessed from the center hospital laboratory. Daily blood glucose levels were assessed by capillary point-of-care (POC) testing pre-breakfast, pre-lunch, pre-dinner and night-time (0300 h) in all patients. A subgroup of patients (n = 69) wore a professional (blinded) Abbott FreeStyle Libre continuous glucose monitor (CGM). Plasma ketone levels were assessed in all patients at randomization (day 0), and on days 3 and 5. All patients with plasma ketone levels ≥ 3.0 mmol/L underwent venous blood gas analysis done to check for acidosis.

Study outcomes

Efficacy outcomes

The primary end point of the study was to determine whether the use of dapagliflozin 10 mg complementary to basal-bolus insulin was noninferior to the use of the basal-bolus regimen alone for glycemic control in the hospital, as measured by mean daily blood glucose levels. Primary and secondary end points relating to glycemic control were assessed during the first 12 days after randomization. We determined differences in mean daily blood glucose in patients admitted to the cardiac surgery service and the proportion of blood glucose readings within target (70–180 mg/dL) before meals. All patients had 4-point blood glucose readings with point-of-care glucometers (POC blood glucose—4 readings, Accu-Chek Instant meter). Glucose values obtained with CGM (in a subset of patients) were analyzed to compare the mean daily blood glucose, percentage of time that the sensor glucose measurement was within the target range (70–180 mg/dL) or above the target range (> 180 mg/dL) or below the target range (< 70 mg/dL). Glycemic variability was determined by the coefficient of variation of glucose values.

Adverse events and safety

An important safety outcome of this study was to determine the number of patients with severe ketonemia and diabetic ketoacidosis (DKA). Severe ketonemia was defined as plasma β-hydroxybutyrate levels ≥ 3.0 mmol/L. All participants who developed severe ketonemia underwent venous blood gas analysis to diagnose DKA. DKA was defined as venous blood gas pH of < 7.3 and/or serum HCO3- of < 15 mmol/L, in the presence of ketonemia and hyperglycemia. We also determined the number of patients with hypoglycemia as defined by the American Diabetes Association, defined as glucose < 70 mg/dL, clinically significant hypoglycemia (< 54 mg/dL), or severe hypoglycemia (< 40 mg/dL) [11]. In addition, glucose values obtained with CGM were analyzed to compare the percentage of time that the sensor glucose measurement was below the target range (< 70 mg/dL) [12]. A set of hospital complications was investigated, including cardiovascular complications (cardiac arrhythmia requiring medical treatment), acute kidney injury (defined as an increment in serum creatinine > 0.3 mg/dL from baseline) [13], moderate pleural effusion/consolidation requiring intervention, sternal wound infections requiring intervention, and mortality [14, 15]. To assess acute kidney injury, serum creatinine concentrations were estimated at admission (baseline) and at days 1, 2, 4 and 6 following surgery.

Statistical procedures

The hypothesis was that cardiac surgery patients with T2D treated with dapagliflozin complementary to basal-bolus insulin or basal-bolus insulin regimen alone will experience similar improvement (non-inferior) in mean daily blood glucose concentrations during the study period. Since this was a treat-to-target study, we tried to maintain glycemia in the prespecified target range (70–180 mg/dL) in all patients by adjusting insulin doses. The glucose-lowering efficacy of dapagliflozin in hospitalized patients was supposed to be reflected by a decrease in the dose of insulin and/or the number of daily insulin injections. We calculated glycemic control from the day of randomization to a maximum of 12 days. Mean daily blood glucose was calculated by taking an average of each participant's daily blood glucose (4 readings) and then an average was taken for all days in the study divided by the number of days in the study. We included blood glucose readings from hypoglycemic events. To show the noninferiority of dapagliflozin plus insulin and the basal-bolus regimen alone in terms of glycemic control, we set the equivalence margin as 18 mg/dL (1 mmol/L) because a difference < 18 mg/dL is usually not considered as clinically significant [1, 16]. Based on the results of previous basal-bolus insulin trials in hospitalized patients with T2D [2, 17, 18], it is reasonable to assume that the standard deviation of mean daily blood glucose is bounded above 45 mg/dL (2.5 mmol/L) [18, 19]. Assuming the true blood glucose difference between the treatment groups is zero, and using one-sided, two-sample t-tests, we estimated that 110 subjects were required for each treatment group to achieve 90% power, with alpha = 0.05. Accounting for < 5% attrition rate, we needed to recruit at least 120 patients per treatment group.

All analyses were performed according to the intention-to-treat (ITT) principle, and included data from all participants who had undergone randomization. To compare continuous baseline and clinical characteristics and outcomes between the two study groups, we used nonparametric Wilcoxon rank-sum tests. For discrete baseline and clinical characteristics and outcomes (such as hypoglycaemia outcomes), we conducted non-parametric comparisons based on a two-sided chi-squared test (or Fisher's exact test). A p value of less than 0.05 was considered statistically significant. The data are presented as the mean ± SD for continuous variables and count (%) for discrete variables. All statistical analyses were performed using the SAS System version 9.4 (SAS Institute, Cary, NC, USA).

Results

Description of study population

Between August 4, 2022, and January 12, 2023, a total of 542 consecutive patients admitted for cardiac surgery were screened for inclusion in the trial. Of these patients, 330 cardiac surgery patients with T2D were eligible for the study. Following exclusion, 250 patients were selected for the study. Of these patients, 125 participants were randomly assigned to receive dapagliflozin 10 mg once daily plus basal-bolus insulin (DAPA group) and 125 to receive basal-bolus insulin regimen alone (INSULIN group). All randomized participants were included in the intention-to-treat analysis. Furthermore, 33 patients in the DAPA group and 36 patients in the INSULIN group were randomly selected for blinded professional continuous glucose monitoring (FreeStyle Libre CGMS). The CONSORT diagram of this study is presented in Fig. 1. The median age of the patients was 61 years (range, 55–67), and 219 (87.6%) were men. With respect to covariates, the data in Table 1 and supplementary table S2 suggest that the two groups were well balanced without significant differences in age, body mass index, duration of diabetes or baseline glycated hemoglobin (HbA1c).

Efficacy outcomes

In this intention-to-treat population, both treatment regimens (dapagliflozin 10 mg once daily plus basal-bolus insulin or basal-bolus insulin regimen) resulted in prompt improvement in mean daily blood glucose concentrations during the hospital stay and a week postdischarge from the hospital (a maximum exposure of 12 days) (Fig. 2). The mean daily blood glucose concentration in the DAPA group (149 [SD 16]) was not inferior to that in the INSULIN group (150 [16]), with a mean blood glucose difference of −1 (95% CI −5, 3) (Table 2). There was no difference in the mean blood glucose concentrations pre-breakfast, pre-lunch, pre-dinner and night-time (0300 h) (Table 2, Supplementary figure S2). There were no differences between mean percentage of blood glucose readings between the target blood glucose of 70–180 mg/dL (82.7% [23.7] vs. 82.5% [24.8], p = 0.816) between the DAPA and INSULIN groups (Table 2). There was no difference in the total daily insulin dose (mean, 39 units [3] vs. 40 units [20] between the DAPA and INSULIN groups. Similarly, there were no differences between groups in total basal and total prandial insulin doses (Table 2).

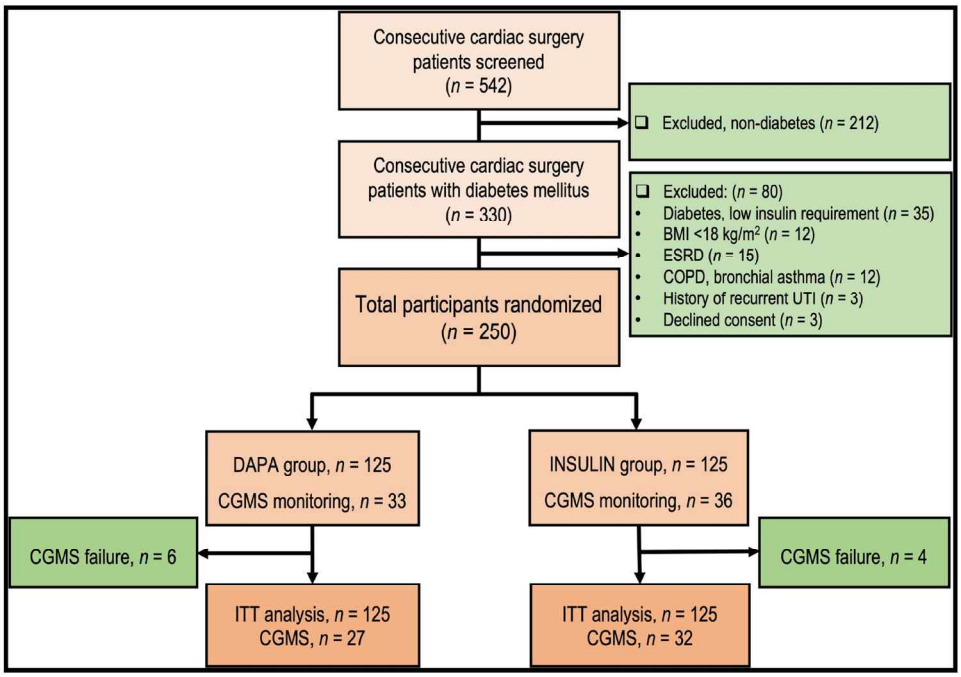


Fig. 1 Consort diagram of the study. BMI body mass index; ESRD end stage renal disease; COPD chronic obstructive pulmonary disease; CGMS continuous glucose monitoring system; ITT intention-to-treat

Table 1 Baseline characteristics of all participants (n = 250)

Parameter	DAPA group (n = 125)	INSULIN group (n = 125)	Total (n = 250)
Median age (IQR), years	59 (54–67)	63 (55–68)	61 (55–67)
Sex, (M/F)	112/13	107/18	219/31
Weight, kg	70.3 (9.8)	70.6 (10.8)	70.5 (10.3)
Body mass index, kg/m^2	25.3 (3.1)	25.7 (3.7)	25.5 (3.5)
Median diabetes duration (IQR), years	6 (4.5–10)	5.2 (4–10)	6 (4–10)
Admission diabetes therapy, n (%)			
Diet only	9 (7.2)	10 (8.0)	19 (7.6)
Oral antidiabetic agents	106 (84.8)	108 (86.4)	214 (85.6)
Oral antidiabetic agent + insulin	10 (8.0)	7 (5.6)	17 (6.8)
Comorbidities, n (%)			
Hypertension	100 (80.0)	97 (77.6)	197 (78.8)
Dyslipidemia	42 (33.6)	52 (41.6)	94 (37.6)
Hypothyroidism	17 (13.6)	15 (12.0)	32 (12.8)
Obesity	61 (48.8)	58 (46.4)	119 (47.6)
Left ventricular ejection fraction (LVEF) at admission			
LVEF ≥ 50% (normal), n (%)	67 (53.6)	78 (62.4)	145 (58.0)
LVEF < 50% (reduced), n (%)	58 (46.4)	47 (37.6)	105 (42.0)
Glycemic variables at baseline			
HbA1c, %, mean (SD)	7.8 (1.4)	7.6 (1.5)	7.7 (1.4)
HbA1c, > 9.0%, n (%)	20 (16.0)	25 (20.0)	45 (18.0)
Admission random BG, mg/dL	157 (50)	155 (53)	156 (52)
Randomization BG, mg/dL	166 (37)	164 (38)	165 (37)
Randomization BG ≤ 180 mg/dL, n (%)	79 (63.2%)	87 (69.6%)	166 (66.4%)
Randomization BG > 180 mg/dL, n (%)	46 (36.8%)	38 (30.4%)	84 (33.6%)

Data are mean (SD), median (IQR) and n (%) as indicated. To convert HbA1c to the SI units of mmol/mol, multiply the HbA1c percentage value × 10.93 and subtract 23.5 from the product. To convert glucose to mmol/L, multiply the value × 0.0555

HbA1c glycated hemoglobin; BG blood glucose; LVEF left ventricular ejection fraction

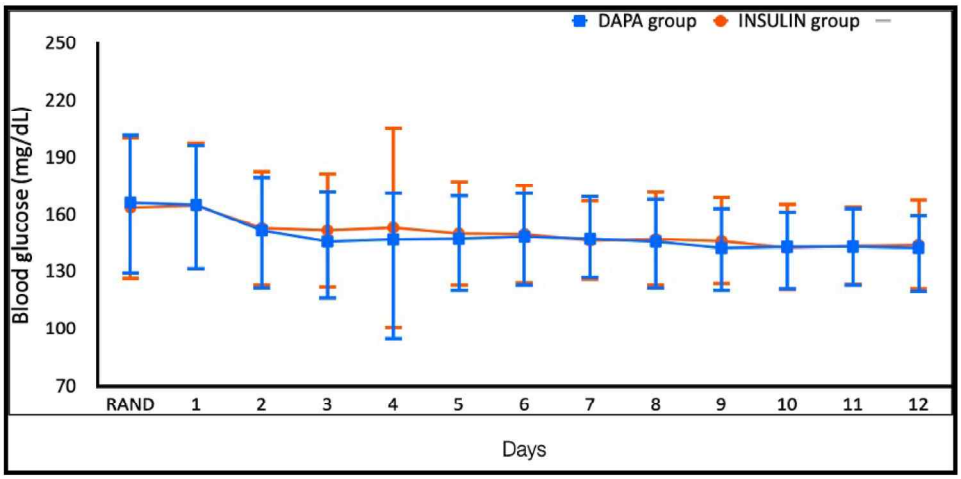


Fig. 2 Daily mean (SD) blood glucose concentrations

Table 2 Glycemic outcomes and insulin doses (POC-BG data analysis)

Parameter	DAPA group (n = 125)	INSULIN group (n = 125)	p value
Glycemic outcomes			
Total number of BG readings	5147	5061	–
Mean daily BG, mg/dL	149 (16)	150 (17)	0.529
Mean daily BG after day 1, mg/dL	147 (16)	148 (17)	0.503
Mean percentage of BG readings between 70 and 180 mg/dL	82.7 (23.7)	82.5 (24.8)	0.816
Mean percentage of BG readings between 70 and 140 mg/dL	45.6 (31.3)	45.7 (32.5)	0.959
Mean percentage of BG readings between 100 and140 mg/dL	41.5 (30.5)	41.4 (30.3)	0.934
Mean percentage of BG readings between 181 and 250 mg/dL	15.3 (21.4)	15.1 (22.1)	0.885
Mean percentage of BG readings > 250 mg/dL	1.7 (7.8)	2.2 (9.0)	0.169
Mean pre-breakfast BG, mg/dL	138 (17)	140 (16)	0.495
Mean pre-lunch BG, mg/dL	157 (23)	160 (29)	0.379
Mean pre-dinner BG, mg/dL	160 (23)	158 (22)	0.694
Mean 0300 h BG, mg/dL	139 (18)	141 (21)	0.553
Insulin dose			
Total daily, units/day	39 (16)	40 (21)	0.450
Total basal, units/day	19 (7)	19 (9)	0.835
Total prandial, units/day	20 (10)	22 (13)	0.271
Total daily, units/kg/day	0.56 (0.26)	0.58 (0.29)	0.598
Number of injections/day, median (IQR)	3.9 (3.8–4)	4 (3.9–4)	0.246

Data are mean (SD) and n (%). To convert HbA_{1c} to the SI units of mmol/mol, multiply the HbA_{1c} percentage value × 10.93 and subtract 23.5 from the product. To convert glucose to mmol/L, multiply the value × 0.0555

POC point-of-care; BG blood glucose

Continuous glucose monitoring measurements

In our study, 69 patients were randomly selected for continuous glucose monitoring, 33 in the DAPA group and 36 in the INSULIN group. The CGM data of 27 patients in the DAPA group and 32 in the INSULIN group were finally available for analysis. The mean daily blood glucose level by CGM in the DAPA group (126 mg/dL [SD 27]) was not inferior to that in the insulin group (132 mg/dL [35]), with a mean difference of −6 mg/dL (p = 0.442). The DAPA and INSULIN groups both spent a similar percentage of time in the target glucose range (76.2% vs. 72.9%, p = 0.453), and had comparable amounts of time above the target range (13.2% vs. 17.4%, p = 0.343), and time below the target glucose range (10.7% vs. 9.6%, p = 0.752) (Supplementary Table 3).

Safety outcomes

There were no differences in the number of patients with blood glucose < 70 mg/dL between groups by capillary point-of-care testing. Clinically significant hypoglycemia (< 54 mg/dL) was infrequent with 6 episodes in the DAPA group and one episode in the INSULIN group (Table 3). The median inpatient stay was 10 days (interquartile range [IQR] 8, 11) in DAPA group patients versus 10 days (IQR 9, 11) in INSULIN group patients (p = 0.704). The total number of complications was also comparable between the two groups (Table 3).

The mean plasma ketone levels were significantly higher in the DAPA group than in the INSULIN group both at day 3 (0.71 vs. 0.30 mmol/L, p < 0.0001) and day 5 (0.42 vs. 0.19 mmol/L, p < 0.0001) of randomization (supplementary figure S3). Six patients (5%) in the DAPA group and no patient in the INSULIN group developed severe ketonemia at day 3. No patient developed diabetic ketoacidosis (Supplementary table S4).

The baseline mean creatinine levels (0.90 vs. 0.96 mg/dL, p = 0.086) were similar between the two groups (Supplementary Fig. 4, Supplementary table 5). Nine patients (7.2%) in the DAPA group and 16 (12.8%) in the INSULIN group developed acute kidney injury (Table 3). However, all patients with acute renal injury improved by the time of discharge from the hospital.

Discussion

In hospitalized patients with T2D admitted for cardiac surgery, treatment with dapagliflozin 10 mg once a day plus basal-bolus insulin or basal-bolus insulin regimen alone in the early postoperative period resulted in similar glycemic control. There was a rapid improvement in glycemic control in both groups, without significant differences in mean daily blood glucose, number and percentage of blood glucose values within the target of 70–180 mg/dL, total daily insulin doses and number of daily insulin injections. As the use of dapagliflozin complementary to basal-bolus insulin did not reduce insulin dose or the number of insulin injections per day, therefore dapagliflozin lacks glycemic efficacy in hospitalized cardiac surgery patients. Furthermore, dapagliflozin increased mean plasma ketone levels significantly both at days 3 and 5. Six patients in the DAPA group developed severe ketonemia compared to none in the INSULIN group.

Multiple insulin injections (basal-bolus insulin) are the standard of care for hospitalized patients in the noncritical care setting. However, basal-bolus insulin is labor intensive. Therefore, oral antidiabetic medication has been examined for the treatment of inpatient hyperglycemia in the noncritical care setting. DPP4 inhibitors (sitagliptin, linagliptin) complementary to insulin have been demonstrated to be safe and improve glycemia similar to basal-bolus insulin [3–6]. In these studies, the glycemic efficacy was indicated by reductions in the doses of insulin and the number of insulin injections, as target glucose levels were maintained similarly

Table 3 Safety outcomes: hypoglycemia (POC-BG data) and hospital complications

Parameter	DAPA group (n = 125)	INSULIN group (n = 125)	P value
Hypoglycemic (safety) outcomes, n (%)			
Patients with any BG reading < 70 mg/dL	12 (9.6%)	9 (7.2%)	0.351
Number (%) of BG readings < 70 mg/dL	13 (10.4%)	9 (7.2%)	0.294
Patients with any BG reading between 54 and 69 mg/dL	7 (5.6%)	8 (6.4%)	0.790
Number (%) of BG readings between 54 and 69 mg/dL	7 (5.6%)	8 (6.4%)	0.790
Patients with any BG readings < 54 mg/dL	5 (4.0%)	1 (0.8%)	0.901
Number (%) of BG readings < 54 mg/dL	6 (4.8%)	1 (0.8%)	0.901
Participants with hospital complications, n (%)			
Length of hospital stay, days	10 (8–11)	10 (9–11)	0.704
Total number of complications	27 (21.6)	31 (24.8)	0.549
Acute kidney injury, n (%)	9 (7.2)	16 (12.8)	0.14
Moderate pleural effusion and/or consolidation requiring intervention	2 (1.6)	5 (4.0)	0.250
Sternal wound infection, requiring surgical intervention	3 (2.4)	2 (1.6)	0.651
Cardiac arrhythmia requiring medical treatment	7 (5.6)	7 (5.6)	1.000
Severe ketonemia, n (%)	6 (4.8%)	0 (0%)	0.013

Data are n (%)

BG blood glucose

in all patients by adjusting insulins. Our study examined the glycemic efficacy of dapagliflozin, an SGLT2 inhibitor, which is another class of oral antidiabetic medication. In our study, both the DAPA and INSULIN groups showed similar improvements in glycemia. However, the use of dapagliflozin did not result in reductions in insulin doses or the number of insulin injections, indicating a lack of glycemic efficacy of dapagliflozin for the hospital management of hyperglycemia in the noncritical care setting. This lack of glycemic efficacy of dapagliflozin in hospitalized patients was also demonstrated by Gerards et al. in a pilot study in patients with prednisolone-induced hyperglycemia during treatment for acute exacerbation of chronic obstructive pulmonary disease compared with placebo [20]. In the study by Gerards et al., blood glucose was monitored by CGMS in all patients (n = 46). The mean glucose concentration was 182 mg/dL in the dapagliflozin group versus 187 mg/dL in the placebo group, which was not a statistically significant reduction (mean difference -5 mg/dL, p = 0.66). In our study, we also monitored glucose levels by CGMS in a subgroup of patients (n = 59). In our study, the mean glucose concentration by CGM was 126 in the DAPA group versus 132 mg/dL in the INSULIN group (mean difference −6 mg/dL, p = 0.44). The reason for the lack of glycemic efficacy of SGLT2 inhibitors in cardiac surgery patients during hospitalization is not clear. One hypothesis could be a lack of adequate action at the kidneys in hospitalized patients. However, we checked urine glucose levels in several patients in both groups during the hospital stay after randomization. Twenty-eight out of 33 patients (84%) had presence of glucose in urine in the DAPA group, while only 4 patients out of 25 patients (16%) had glucosuria in the INSULIN group, indicating intact action of dapagliflozin at the kidneys. The lack of glycemic efficacy of dapagliflozin was consistent across several subgroup analyses, including patients with both normal and reduced left ventricular ejection fraction (LVEF < 50% vs. ≥ 50%). Further research should focus on the reasons for the lack of glycemic efficacy of SGLT2 inhibitors in hospitalized patients.

SGLT2 inhibitors are increasingly being used in hospitalized patients due to multiple pleiotropic effects [7]. These drugs reduce cardiovascular mortality, especially by reducing the risk of heart failure, and improve renal outcomes [7]. Recently, two randomized controlled trials demonstrated improvement in several cardiac outcomes when SGLT2 inhibitors (empagliflozin and dapagliflozin) were initiated in patients hospitalized for acute heart failure, with or without diabetes [8, 9]. In a retrospective analysis of heart failure patients (INSIGHT-HF study), inpatient initiation of an SGLT2 inhibitor was associated with a significantly higher postdischarge rate of SGLT2 inhibitor prescriptions and significantly lower heart failure readmission rates at 30 days [21]. Even if SGLT2 inhibitors lack inpatient glycemic efficacy, these agents will still be prescribed increasingly during hospitalization because of cardiac benefits. Therefore, it is important to examine the safety profile of these antidiabetic agents in the inpatient setting. The most important safety concern of SGLT2 inhibitor use in metabolically challenged patients is the occurrence of DKA, especially euglycemic DKA. Current recommendations do not allow SGLT2 inhibitors in hospitalized patients. The endocrinological practice guidelines recommend discontinuing treatment with SGLT2 inhibitors at least 24 h before elective surgery [22]. These guidelines are based on early evidence of an increased incidence of DKA, especially normoglycemic DKA with SGLT2 inhibitor use during metabolically challenging activities [23, 24]. Our study showed an increase in mean plasma ketone levels with dapagliflozin use. Severe ketonemia was observed in six patients, but none developed DKA. During subsequent monitoring, severe ketonemia resolved in all patients despite the continuation of dapagliflozin therapy. Similar to our study, several RCTs of SGLT2 inhibitors for hospitalized patients do not demonstrate an increased occurrence of DKA [8, 9, 25]. In a large RCT (DARE-19), among the 625 COVID-19 patients randomized to the dapagliflozin group, there were only two cases of mild DKA [25]. The reason for the increase in serum ketone levels due to SGLT2 inhibitors is the shift in substrate utilization from carbohydrate to fat oxidation along with an increase in glucagon levels, dehydration and insulinopenia. This along with the precipitating factors that are often present in hospitalized patients, including severe infections, prolonged starvation or surgical interventions increase the risk of DKA. From a pathophysiological standpoint, the development of DKA requires the presence of absolute or relative insulinopenia. Since all our patients were on exogenous insulin therapy, this may also explain the absence of DKA in our study. Moreover, none of our patients were on exclusive insulin therapy prior to the surgery, indicating the absence of insulinopenic patients in our cohort.

As patients hospitalized for cardiac surgeries are often aggressively treated with diuretics and other vasoactive agents, it was possible that an SGLT2 inhibitor would increase the risk of volume depletion and acute kidney injury. In our study, dapagliflozin did not increase the incidence of acute kidney injury compared with basal-bolus insulin. In our study, acute kidney injury occurred in 7.2% of patients in the DAPA group and 12.8% of patients in the INSULN group. In the study by Voors et al., acute kidney injury occurred in 7.7% of patients in the empagliflozin group and in 12.1% of patients in the placebo group [8]. In another hospitalized heart failure trial, 4 patients (10%) in the empagliflozin group and 3 patients (8%) in the placebo group developed worsening kidney function (26). All these

studies, and our study indicate that SGLT2 inhibitors are safe to initiate in hospitalized patients as far as renal function is concerned.

Strengths and limitations

Our study has several strengths. We know from previous studies that dapagliflozin improved cardiac outcomes in hospitalized patients with acute heart failure. We also know from basic studies that SGLT2 inhibitors improve cardiac outcomes when initiated in an ailing heart. Therefore, in the current study, we selected cardiac surgery patients with T2D, so that we could assess glycemic efficacy during hospitalization and use SGLT2 inhibitors in high-risk cardiac patients who might benefit from this therapy. We performed 4-point point-of-care testing in all patients, including three usual premeals (prebreakfast, prelunch and predinner); and one night-time (0300 h). This enabled us to assess nocturnal glycemia. Another strength of our study is that a subset of our recruited patients wore a professional, blinded CGMS, making it possible to assess the rate of nocturnal and asymptomatic hypoglycemia.

Our study has several limitations. Participants and investigators were not masked to treatment in this trial (open-label). We included patients undergoing cardiac surgery, and the results might not be generalisable to all hospitalized patients, especially patients admitted to medical services. Another limitation of our study was that it was not powered to determine differences in complications between groups, but overall rates of hypoglycemia and complications were very low.

Conclusion

In conclusion, treatment with dapagliflozin 10 mg daily plus the basal-bolus insulin regimen, titrated daily to target blood glucose of 70–180 mg/dL, resulted in comparable mean daily blood glucose concentrations compared with the basal-bolus insulin regimen alone in cardiac surgery patients with T2D, in the early postoperative period. The addition of dapagliflozin did not lead to reductions in total daily insulin doses or the number of daily insulin injections, indicating a lack of glycemic efficacy in hospitalized patients. Furthermore, dapagliflozin use lead to severe ketonemia in six patients, however none developed DKA. Further studies are required to look for the safety of dapagliflozin in the hospital setting.

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Author contributions MSK was responsible for the study concept and design, data collection, interpretation of data, drafting of the manuscript, and approval of the final submission. MSK, SKM, PK, MM, AS, CK, and PK contributed to the data collection, insulin dose management, CGM application, and approval of the final submission. MKS and PS contributed to the statistical analysis, critical revision of the manuscript, and approval of the final submission. PK, JSW, HKG, GB, VR, SS and RJ contributed to the data collection, critical revision of the manuscript, and approval of the final submission. MSK is the guarantor of this work and, as such, had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. MSK has received speaker honoraria from Boehringer-Ingelheim, Novartis India, Sanofi, Novo Nordisk, Intas Pharmaceuticals Ltd, Sun Pharmaceutical Industries Ltd, USV Private Limited, and MSD Life Sciences Foundation. PK has received speaker honoraria from Novo Nordisk, Boehringer Ingelheim, and Sanofi. HKG has received speaker and consultant fees from Novo Nordisk and Sanofi. JSW has received speaker honoraria from Abbott India Limited, Novartis, AstraZeneca, and Sanofi. RJ has received speaker honoraria from Sanofi, Novartis, Intas Pharmaceuticals, and Lupin Limited. SKM has received speaker honoraria from Novo Nordisk, Boehringer Ingelheim, Novartis, Sanofi, Lupin Limited and Abbott India Limited.

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Data availability The dataset for the current study is available on reasonable request by contacting MSK.

Declarations

Conflict of interests No other conflicts of interest relevant to this article were reported

Ethics approval and consent to participate The study protocol and informed consent form were approved by the Institutional Review Board and the Ethics Committee.

Informed consent Written informed consent was obtained from all study participants in accordance with the guidelines of the Declaration of Helsinki.

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